## MILITARY SPECIFICATION

# MOUNT, ANTIOSCILLATION SIGHT: 10511004

### 1. SCOPE

1.1 This specification covers one type of mount known as Mount, Antioscillation Sight: 10511004 which supports the Antioscillation Sight: XM55.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

**SPECIFICATIONS**

MILITARY

MIL-F-13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Material, Packaging of

MIL-I-45208 — Inspection Requirements, General Specification for

**STANDARDS**

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

**DRAWINGS**

MUNITIONS COMMAND

F10511004 — Mount, Antioscillation Sight: 10511004

**PACKAGING DATA SHEET**

MIL-P-14232/P10511004 — Packaging of Mount, Antioscillation Sight: 10511004

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

### 3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 **Fabrication.** The mount shall be manufactured in accordance with Drawing F10511004 "Mount, Antioscillation Sight: 10511004" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure the inherent quality and the performance capabilities.

3.3 **General specification.** The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:



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(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

**3.4 Environmental.**

3.4.1 *Vibration.* The mount shall be secured on a vibration fixture that suspends the body of the mount from a horizontal mounting surface. The azimuth lock shall be secured with a torque of 12 inch pounds. The mount shall then be vibrated for five minutes, plus or minus 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second. Subsequent to vibration the mount shall show no evidence of physical failure, damage or loose parts, and shall meet the requirements of 3.5.1 to 3.5.3 inclusive (see 6.3).

**3.5 Performance.** The mount shall be positioned similar to the position required in 3.4.1.

3.5.1 *Azimuth rotation.* With the azimuth wing key (lock) released (counterclockwise rotation) the sight support frame shall rotate through an excursion of 40 degrees to the right and 40 degrees to the left of the central position.

3.5.2 *Azimuth lock.* With the azimuth wing key (lock) secured with the torque of 12 inch pounds the mount shall not move in azimuth when a force of five pounds is applied at the outer edge of the frame.

3.5.3 *Polaroid ring sight.* The ring of the sight shall be centrally positioned in the frame assembly. There shall be no blemishes or imperfections that would affect observation through the entire sight or adversely affect image quality (see 6.3).

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 General requirements.** Except where otherwise specified herein, the contractor's inspection system shall be in accordance with MIL–I–45208.

**4.3 Lot formation.** When inspection lot sizes and lot formations are applicable, they shall be in accordance with MIL–STD–105 and shall also be defined in the contractor's inspection system.

4.3.1 *Defective units and lots.* Defective units and lots shall be processed as specified in MIL–STD–105 and shall not be included as a part of the quantity specified in the contract until formal approval is obtained.

**4.4 Inspection provisions.** Components and subassemblies shall be examined in accordance with the inspection provisions contained in MIL–F–13926 and 4.4.1 herein. Test and control of materials not covered by Supplementary Quality Assurance Provisions (SQAP's) or Section 4 of MIL–F–13926 shall be in accordance with "Test Data" provisions of MIL–I–45208. Inspection related to Section 3 of this specification shall be performed in accordance with 4.4.2. The tabulated Classification of Defects listed herein shall constitute the minimum inspection to be performed by the contractor prior to Government acceptance. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

4.4.1 *Inspection of components.* All posi-



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tional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of mount shall be inspected in accordance with the written inspection system required MIL–I–45208. These characteristics are generally defined in MIL–F–13926 as Class 1 or Class 2 Defects.

4.4.1.1 *Component sampling inspection.* Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with the sampling plan tables specified in MIL–F–13926.

4.4.1.2 *Contractor's inspection for all components not covered by SQAP.* Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by MIL–I–45208 and the sampling plan tables MIL–F–13926. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the mount. In addition, the list of Class 1 Defects in MIL–F–13926 shall apply.

4.4.1.3 *Disposition of nonconforming product.* Disposition of nonconforming units shall be in accordance with MIL–STD–105.

4.4.2 *Classification of defects and sampling plans.* Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with MIL–STD–105 and the AQL's specified herein, the classification of defects and sampling plans specified in table I may be used in lieu of 100 percent inspection. The AQL value specified shall apply to each individual characteristic on a single defect basis.

TABLE I. *Classification of defects*

Use inspection level II in table IIIA with sampling plan table IVA of MIL–STD–105.

| Critical: None defined<br>Major: AQL 1.0 percent defective | Requirement paragraph | QA. paragraph | Test method | |
|---|---|---|---|---|
| 101. Physical failure | 3.4.1 | 4.6.1 | Standard (SME) | Measuring Equipment |
| 102. Insufficient azimuth rotation | 3.5.1 | 4.6.2.1 | SME | |
| 103. Azimuth lock failure | 3.5.2 | 4.6.2.2 | SME | |
| 104. Sighting off center | 3.5.3 | 4.6.2.3 | SME | |
| 105. Improper image quality | 3.5.3 | 4.6.2.3 | Visual | |
| Minor: None defined | | | | |

4.4.3 *Control tests.* One mount shall be selected at random from each 50 produced, or, from each month's production, whichever occurs first. Control samples shall be subjected to the examinations and tests specified herein to determine compliance with 3.1, 3.2, 3.3 and 3.4.1. The examinations and tests shall be conducted by the contractor under surveillance of the Government representative at the contractor's facility, or Government approved laboratory, using test equipment authorized for use by the contract or otherwise approved by the contracting officer.

4.4.3.1 *Control sample failure.* Should a control sample fail to meet the requirements specified in 4.4.3, the contractor shall cease those operations causing rejection until necessary corrections have been made by the



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contractor and approved by the Government. When production has resumed, inspection of those requirements that previously caused rejection shall be performed on a 100 percent basis until 5 consecutively produced mounts meet these requirements. When this is accomplished, control inspection (4.4.3) may be continued.

**4.5 Inspection equipment.** Supply, calibration, maintenance and use of inspection equipment shall be in accordance with MIL-I-45208

4.5.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.5.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.5.2 *Inspection equipment design requirements.*

4.5.2.1 *Equipment to test vibration requirements.* The equipment shall consist of a vibration machine capable of providing and regulating the specified amplitude in 3.4.1 within plus or minus .010 inch. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in a suspended position.

**4.6 Test methods and procedures.**

4.6.1 *Vibration.* This test shall be conducted in accordance with Procedure V of MIL-F-13926 except that the time period shall be as specified in 3.4.1. The azimuth lock shall be secured with standard measuring equipment. Subequent to vibration, the mount shall be subjected to a visual and tactile examination and shall show no evidence of damage. The mount shall then be subjected to the tests of 4.6.3.1 to 4.6.3.3 inclusive and shall meet the specified requirements.

4.6.2 *Performance.* The mount shall be positioned as indicated in 3.5 on a device that suspends the body of the mount from a horizontal mounting surface so that the azimuth axis of rotation is vertical (normal operating position). This device shall be leveled by means of a precision 10 second bench level.

4.6.2.1 *Azimuth rotation.* With the mount positioned as specified in 4.6.3, release the azimuth lock Rotate the sight support frame 40 degrees to the right and 40 degrees to the left of the center position to conform to the minimum requirements of 3.5.1. Angular displacement shall be measured with standard measuring equipment.

4.6.2.2 *Azimuth lock.* Utilize the setup as in the previous test. With the azimuth lock secured, and the torque applied as specified in 3.5.2, there shall be no movement of the mount in azimuth when the specified torque value is applied at the outer edge of the frame. The measured torque values shall be applied with standard measuring equipment.

4.6.2.3 *Polaroid ring sight.* Standard measuring equipment shall be used to measure the central position of the ring in accordance with 3.5.3. The optical element shall be examined to determine conformance to requirements listed on the drawing for the polaroid ring sight. The sight shall be subjected to a visual examination for blemishes or imperfections that would impair optical performance, or permit distortion.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging, packing and marking.** Packaging, packing and marking shall be in

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accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10511004. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 **Intended use.** This mount is primarily intended as a mounting device for the Anti-oscillation Sight XM55 when suspended from the overhead section of a helicopter. The sight is used by the gunner/copilot to observe the target and to hold the helicopter on a collision course with the target during the time the missile, launched from the helicopter, is in flight to the target.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1)

(c) Applicable packaging data sheet number (see 5.1)

(d) Applicable stock number

6.3 **Component parts.** When component parts of the mount such as the Polaroid Ring Sight and the Elevation Lock Assembly and Elevation Knob are not fixed to the mount but are installed only when the XM55 Anti-oscillation Sight is installed in the mount. these components shall not be subjected to the requirements and tests of this specification unless specifically identified for same.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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